

Integration of solar PV, fuel cell, and hydrogen storage systems for sustainable desalination: techno-economic and multi-criteria decision-making analysis

Olubayo Babatunde^{a,b,*}, Oludolapo Olanrewaju^b

^a Department of Industrial Engineering, Durban University of Technology, Durban 4000, South Africa

^b Institute of Systems Science, Durban University of Technology, Durban 4000, South Africa

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ABSTRACT

Water scarcity affects approximately 2.1 billion people globally, with remote communities in sub-Saharan Africa particularly vulnerable to inadequate portable water access and related health challenges. While renewable energy integration with desalination has gained attention, comprehensive multi-criteria assessments across multiple coastal locations remain limited. This study presents a techno-economic and multi-criteria optimization of solar PV/fuel cell/battery/electrolyzer/hydrogen storage systems for powering seawater desalination plants in four South African coastal provinces. Using the Hybrid Optimization Model for Electric Renewables (HOMER) software, system sizing, economic analysis, reliability assessment, and environmental impact evaluation were conducted through energy-balance simulations, followed by entropy-weighted multi-criteria decision-making analysis using WSM, WASPAS, EDAS, and ARAS methods. The results show that Alexandra Bay emerged as the optimal location, achieving the lowest total net present cost (\$122,742), levelized cost of energy (\$0.242/kWh), levelized cost of water (\$0.796/m³), highest essential load reliability (100 %), and minimal environmental impact (3.47 kg/yr NO_x emissions, 5,504 L/yr water consumption). However, fuel cell capacity factors across all locations remained extremely low (0.38–0.84 %), with hydrogen production significantly exceeding consumption, suggesting that battery storage alone may suffice for this application scale. The results demonstrate that solar resource availability and dispatch strategy selection significantly influence system performance, providing a robust framework for sustainable water-energy nexus solutions in water-scarce coastal regions. Finally, the economic viability of hydrogen-integrated desalination systems will be increased if the capital costs associated with fuel cells and hydrogen storage are reduced by 60–80 % below current values.

Introduction

The provision of fresh portable water and reliable energy is essential in the development and sustainability of any society [1]. While fresh water is needed for survival in terms of food production, drinking, irrigation, animal husbandry, and industrial activities, reliable energy (especially electricity) is essential for the running of the processes that leads to these activities. However, with the continuous increase in the global population comes the pressure on the available natural water and energy resource. The pressure on the energy and water sectors is due to rise in industrialization, agricultural activities, and other economic development activities, all because of population growth [2].

It is reported that globally, approximately 2.1 billion people lack

access to adequate portable water [3]. The pressure from accessibility to fresh water negatively affects both the availability and the quality of portable water supplies. At present many parts of the world are faced with acute water scarcity with a high tendency of precipitating drought and consequently famine. In many remote communities, especially in sub-Saharan Africa, many diseases, and deaths (especially among children) are related to water scarcity or water which are unfit for human consumption [4]. Apart from the health challenges associated with water scarcity or inadequacy of portable water, it could also lead to reduction in food and animal production. Since approximately 97 % of the total global water resources is saline, the obvious option to accessing sustainable freshwater is through desalination of seawater and water recycling [5].

* Corresponding author.

E-mail address: babatundeom@ieee.org (O. Babatunde).

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Based on the technology adopted, there are two main methods by which desalination can be achieved, these include thermal desalination and membrane desalination technologies [6]. The thermal desalination technology is based on phase change- a principle by which the salt water is boiled and evaporated to obtain a condensate in form of water vapour and salt as the by product. The water vapour obtained in the process is a pure water free of salt. Some of the most common principles used in thermal desalination include vapour compression evaporation, solar water desalination, multi-stage flash distillation, multi-effect distillation and cogeneration. In the membrane desalination, relatively permeable membranes are used to move either salt or water to obtain two zones of varying concentrations to supply fresh water. The membrane desalination techniques are classified as electro-dialysis, membrane distillation and reverse osmosis. The most common of the membrane desalination methods is reverse osmosis. It is a process which is based on pressure difference, in which a salt solution disperses its solutes by diffusion across a membrane [7].

Although, seawater purification can offer a climate-neutral source of portable water; however, most of the common methods of purification are usually not environmentally sustainable and energy-intensive [8]. If the process of desalination is to be sustainable, the water obtained from such process must be economically sustainable, the processes involved in the purification must be environmentally friendly and most importantly, the source of energy for the desalination plant must be sustainable. For this reason, it is important to power the desalination plant using alternative source of energy such as solar and wind. The use of any of these energy systems would depend on the availability of solar and wind resource in the area where such system is to be installed.

The integration of desalination and hydrogen production within a renewable power framework aligns with the emerging Power-to-X (P2X) concept, which promotes the conversion of renewable electricity into multiple value chains beyond direct power supply. In this context, electricity generated from solar PV is utilized not only for water production but also for hydrogen generation through electrolysis, thereby linking energy and water systems in a complementary manner. This hybrid arrangement functions as a Power-to-Hydrogen-to-Water pathway in which electricity serves both as an input for freshwater generation and as a driver for hydrogen storage. Such coupling enhances resilience and optimization of renewable resources, supports off-grid resilience, and provides an operational pathway toward sectoral integration within the energy–water nexus.

In practical applications, the intermittency of renewable resources often limits the stability of single-source energy systems; therefore, hybridization and the inclusion of energy-storage components remain essential for ensuring reliability and cost effectiveness[9,10]. Within the Power-to-X context, energy storage, particularly through hydrogen acts as a bridge that enables continuous operation of desalination facilities even when solar or wind resources fluctuate. The broader integration of renewables into desalination systems also offers significant environmental advantages by reducing or eliminating greenhouse-gas emissions typically associated with fossil-fuel-based plants.

Furthermore, recent reports indicates that about 99 % of the total desalinated water are still powered using conventional sources of energy while only 1 % are powered using renewable energy [11], underscoring the urgent need for cleaner alternatives. Ensuring sustainability therefore requires that the energy resources powering desalination plants be locally sourced[12]renewable[8], and economically maintainable by the host communities. In this regard, adopting renewable-driven, hydrogen-integrated desalination aligns with the Power-to-X strategy and advances progress toward Sustainable Development Goals 6, 7, 11, and 13. For a water-scarce coastal nation such as South Africa, bordered by abundant seawater and rich solar potential, the prospect for renewable-energy-powered desalination systems is particularly high, presenting an opportunity for sustainable freshwater generation through the Power-to-Hydrogen-to-Water pathway.

The main objective of this study is to evaluate an integrated Power-

to-Hydrogen-to-Water system that couples solar photovoltaic, fuel cell, and hydrogen storage technologies for seawater desalination in South Africa. The analysis employs an energy-balance approach in HOMER to assess the technical performance, energy flows, and conversion efficiencies of the hybrid configuration, while a multi-criteria decision-making (MCDM) approach is applied to rank alternative system options based on techno-economic and environmental criteria. Through this dual framework, the study aims to demonstrate how renewable electricity can be effectively converted into both clean water and storable energy, thereby enhancing reliability, reducing emissions, and advancing sustainable development objectives consistent with the Power-to-X paradigm. The energy-balance simulation approach used in HOMER is based on the principle of energy conservation, ensuring that power generation equals demand plus losses at each hourly time step. HOMER applies this as a sequential physical rule while simulating all feasible system configurations and ranking them by Total Net Present Cost (TNPC). Unlike Mixed-Integer Linear Programming (MILP) models, which formulate energy balance as a mathematical constraint in a global cost-minimization problem, HOMER performs enumerative, derivative-free optimization that captures nonlinear efficiencies and realistic dispatch behavior. This approach provides a transparent and computationally efficient means for comparative techno-economic analysis across multiple sites (Lambert et al., 2006; Lilienthal et al., 2005; Bahramara et al., 2016).

The rest of this paper is organized as follows: Section 2 presents a comprehensive literature review of renewable energy-powered desalination systems, highlighting research gaps and positioning this study's contributions. Section 3 details the methodology, including site description of the four coastal South African locations, desalination plant specifications, renewable energy system modeling, economic analysis framework, reliability assessment methods, and multi-criteria decision-making approach. Section 4 provides comprehensive results including optimal system configurations, techno-economic performance, operational dynamics, reliability assessment, water consumption analysis, and multi-criteria decision-making rankings for all four locations. Finally, section 5 concludes the study with key findings, policy recommendations, and suggestions for future research directions.

Literature review

The subject of integration of water desalination and renewable energy has been extensively studied in recent times. Various authors have presented case studies, scenarios, and theoretical insights for the adoption of renewable energy for powering water desalination plants. For instance, to power a desalination plant that would be sited in Baltim, Egypt, researchers presented a feasibility study into the adoption of a hybrid pumped-hydro, battery, diesel generator, wind, solar PV energy system[13]. The work further presented a selection framework based on Fuzzy-AHP and Fuzzy-VIKOR to identify the best option based on 10 performance attributes. The result of the study shows that the most preferred alternative is a combination of 328 kW solar PV array, 5 number 20 kW wind turbines, 100 kW diesel generators with the battery bank consisting of 112 batteries. Though the work presented a robust analysis, important technical details, and attributes such as reliability, excess energy production and total energy production were neglected. In similar research,[14]proposed fuel cell as an alternative for battery storage in the adoption of a solar PV energy system for the powering of a water desalination plant in Minya city, Egypt. Result show that to pump 75 m³ of water daily, it requires the combination of a 66 kW solar PV array, 9 kW fuel cell, 70 kg hydrogen tank, 15 kW electrolyzer and 25 kW converter. The authors of the paper did not include the issue of sustainability from the technical, environmental, and economic perspective in selecting the PV/SCFC HRES as the best as compared to grid extension. The comparative technical viability and cost implication of deploying 6 different types of off-grid hybrid renewable energy systems to power a desalination plant in South Khorasan, Iran has been

investigated using improved bees algorithm[15]. Results of the study show that it is cheaper to deploy battery bank as energy storage device rather than, the hydrogen energy storage. The selection of the optimal system was based on a single criterion which is insufficient for robust sustainability. A hypothetical study that explores the use of a solar microgrid to supply power to a desalination plant in El Paso, USA under various tracking conditions has been presented[16]. It is reported that the optimal system in terms of cost reduction is one that would use one axis tracker in harvesting maximum power. Unlike the study conducted by in Egypt[13], the one conducted in El Paso did not consider the subject of sustainability. To support the Paris agreement in the reduction of GHG emissions,[17] presented a practical case study in which a desalination plant located in Sinai Peninsula was powered using hybrid renewable energy system. The result of the study shows that a PV, battery system hybridized with diesel generator is found to be the most cost-effective, sustainable, and socially acceptable alternative. Although the authors reported that the optimal choice was sustainable, it was only selected based on the economic attribute. For sustainability to be attained and assured, other factors related to technical, environmental and social perspectives must be included [18].

In another paper, a techno-economic assessment of an energy system aimed at powering a desalination plant with a daily capacity of 10 m³ was presented [19]. The result of the study shows that it is more economical to operate the desalination plant on a PV system with a battery back-up rather than using a diesel generator as a back-up.[7] assessed the operations of 7 distinct grid-independent energy system for powering the load demand of a small-scale reverse osmosis system and concluded that water purification could be cost effective for desalination plants sited at areas with rich solar and wind resource. A techno-economic feasibility comparison of powering a water desalination plant based on electrodialysis reversal and reverse osmosis technologies with solar PV although shows technical viability of both technologies, at higher water salinity, it is more economical to use the RO technology [20]. In addition to both sizing and economic assessment of a PV system aimed at powering a desalination plant,[21]assessed the reliability of the proposed system. It further ranked the proposed locations based on data envelopment analysis. In their own contribution,[22]presented a cost and size optimization framework for a energy system that is intended to supply a desalination plant with a capacity of 50 m³ per day. Furthermore, the efficacy of deploying a hybrid energy system consisting of PV, battery, diesel generator and hydrokinetic turbine has been presented [23]. This energy system configuration shows the relevance of water-energy nexus and is applicable in places where the river flow strong enough to drive the turbine. An analysis performed by [24]shows that for a location in southeast of Egypt, a hybrid photovoltaic/wind/diesel/battery energy system outperformed the other feasible systems in terms of costs and emission reduction. In a more recent study, a Fuzzy-TOPSIS MCDA with five attributes was adopted in the ranking of energy alternatives used for satisfying desalination plant electricity demands for some coastal cities in India[25].

The techno-economic and environmental impact of adopting renewable energy power plant in place of existing fossil-fuel powered energy system for desalination plants in the Adrar region of Algeria has also been investigated[26]. The study considered the use of fossil fuel only, renewable energy systems only and a mix of both renewable and fossil fuel. The proposed renewable energy-based RO system achieves a 59 % cost reduction while integrating 10.34 kW of PV power reduces diesel generator capacity by 48 %, resulting in 58 % fuel savings and 62 % lower CO₂ emissions compared to conventional systems. Furthermore, Qudah et al. optimized hybrid PV-wind-water tank storage systems for RO desalination in Saudi Arabia, achieving \$0.95/m³ water cost using surplus energy management instead of expensive batteries [27].

It is evident from the literature that extensive research efforts have been dedicated to the subject of integrating renewable energy into powering desalination plants. All the studies reviewed in this paper considered the technical sizing and the economic implications of the

renewable energy system that is needed to power desalination plants while majority of them considered the environmental impact of the proposed energy system (Table 1). However, only few studies have considered the reliability of the proposed energy system as well as the issues of multiple criteria in the ranking of the system alternatives using MCDA. These two themes (reliability and multicriteria analysis) are particularly very important for decision making. In addition to evaluating the techno-economic and environmental impact assessment of energy system for powering a desalination plant, this study evaluates the reliability of the system alternatives and rank them based on MCDA.

Contribution to Knowledge

The concomitant inclusion of the sizing, economic, environmental impact, reliability, and multi-criteria ranking of feasible system alternatives in a single study serves as the main contribution of this study. Also, based on the knowledge of the authors, this is the first paper that presents a case study of deploying hybrid renewable energy for powering water desalination plants in the four coastal provinces of South Africa. This study advances the literature on renewable-powered desalination systems by moving beyond routine software-based techno-economic analysis and offering new methodological insight, system-level understanding, and decision-oriented evidence for hydrogen-integrated water-energy systems.

First, the study introduces a hybrid energy-water optimization framework that explicitly integrates energy-balance simulation, reliability risk metrics, and entropy-weighted multi-criteria decision analysis within a single decision pipeline. While HOMER is used as the simulation engine, the contribution lies not in the software itself but in how its outputs are extended and reinterpreted. Unlike conventional HOMER-based studies that rely primarily on cost minimization, this work operationalizes Primary Load Ratio (PLR), Energy Not Supplied (ENS), and Loss of Power Supply Probability (LPSP) as system-wide reliability indicators that jointly capture the interaction between essential desalination loads and flexible hydrogen production. This dual-load reliability formulation is rarely applied in desalination studies and provides a more realistic representation of Power-to-Hydrogen-to-Water systems than standard unmet-load metrics. The work also contributes new system-level understanding by critically examining the functional role of hydrogen within small-scale renewable desalination architectures. Rather than presuming hydrogen storage as inherently advantageous, the framework evaluates its operational relevance in relation to dispatch behavior, storage interaction, and competing battery-based alternatives. This analytical treatment enables a principled assessment of when hydrogen integration is structurally justified and when simpler storage configurations may be more appropriate, thereby refining prevailing assumptions in the literature.

In addition, the study advances decision-making practice by combining objective entropy-based weighting with multiple ranking algorithms (WSM, WASPAS, EDAS, and ARAS) to test ranking robustness across methodological perspectives. Rather than relying on a single multi-criteria method, the study cross-validates system rankings under additive, multiplicative, distance-based, and ratio-based logics. This approach reduces ranking bias and strengthens confidence in location-specific recommendations, addressing a key weakness in prior desalination studies that depend on single-method MCDA or subjective weighting schemes. The study further contributes empirically by offering a geographically comparative framework that evaluates renewable desalination systems across multiple coastal regions under consistent system assumptions. This approach isolates the influence of resource availability, system configuration, and dispatch strategy on sustainability performance, addressing a gap in prior studies that are typically limited to single locations cases.

Finally, the study extends the theoretical framing of renewable desalination by explicitly positioning the system as a Power-to-Hydrogen-to-Water pathway within the broader Power-to-X paradigm.

Table 1

Reference	location	Feed	Production capacity	Energy demand	Energy configuration	Themes				
						Technical	Economic	Environmental	Reliability	MCDA
[13]	Egypt	Seawater	1000 m ³ /day	2133.2 kWh/day 333.15 kW peak	WD/PV/DG/BAT	✓	✓	✓		✓
[14]	Egypt	brackish water	150 m ³ /day	236 kWh/day	PV/FC	✓	✓	✓		
[15]	Iran	NS	NS	4 kWh/m ³	WT/HS PV/WT/HS PV/HS WT/BAT PV/WT/BAT PV/BAT	✓	✓		✓	
[16]	USA	brackish water	104,000 m ³ /day	623.75 kW/day	Solar PV with two axis trackers, net metering	✓	✓			
[17]	Egypt	brackish water	100 m ³ /day	557.22 kWh/day	PV/DG/BAT	✓	✓	✓		
[19]	Oman	Seawater	10.4 m ³ /day	60.3 kWh/day	PV/DG/BAT	✓	✓	✓		
[7]	Turkey	Seawater	1 m ³ /hour	4.38 kWh/m ³	WD/PV/DG	✓	✓	✓		
[21]	Iran	Seawater	228 m ³ /day and 148 m ³ /day	0.83 kWh/m ³	PV/BAT	✓	✓		✓	
[22]	Spain	Seawater or Brackish water	50 m ³ /day	250 kW h/day	WD/PV/DG	✓	✓	✓		
[23]	Egypt	Seawater	1 m ³ /h	4.38 kWh/m ³	WD/PV/DG	✓	✓	✓		
[24]	Egypt	Seawater	1500 m ³ /d	4393 kWh/day	WD/PV/DG/BAT	✓	✓	✓	✓	
[25]	India	Seawater	1.5 m ³ /h	NS	WD/PV/HKT/BAT	✓	✓	✓		✓
[26]	Adrar, Algeria (oil refinery)	Industrial water (for RO desalination)	30.0 m ³ /h	17.5 kW RO plant	Three scenarios: (1) PV + Battery only, (2) PV + Battery + Diesel hybrid, (3) Diesel only	✓	✓	✓	✓	
[27]	Saudi Arabia	Seawater	5 m ³ /day (constant) / 15 m ³ /day (variable)	Variable RO capacity	PV + Wind Turbine + Water Tank storage (no batteries)	✓	✓	✓		
[37]	Egypt	Brackish water	65 m ³ /day	6.6 kW system power	Six configurations: PV + Battery, WT + Battery, PV + WT + Battery, PV + WT + DG, PV + DG + Battery, PV + WT + DG + Battery	✓	✓	✓		
[38]	Algeria	Seawater	90,000–300,000 m ³ /day	21.5–101.3 MW total	Solar PV + RO desalination + PEM electrolysis + battery storage	✓	✓	✓	✓	
Present study	South Africa	Seawater	10 m ³ /day		PV/WD/DG/BAT/FC	✓	✓	✓	✓	✓

This conceptual linkage clarifies how renewable electricity can simultaneously support freshwater production and energy storage, while also identifying efficiency losses and economic bottlenecks that must be addressed for practical deployment. In doing so, the work contributes to a more nuanced understanding of sector coupling at the water–energy nexus.

Methodology

This section presents the methodology adopted in this study. The main goal of this study is to propose an optimal energy system (based on the available energy resource) for powering a hypothetical desalination plant costal in major coastal cities in South Africa. The desalination plant is expected to provide freshwater in scarce areas of the provinces that have borders with the sea in South Africa. The proposed system is expected be modular, scalable, and mobile do that it could be deployed to any part of the country for disaster relieve. In identifying the optimal systems, three major themes of sustainability are considered, they

include technical, economic, and environmental implication of adopting the energy systems. By implication, the proposed energy system must be reliable, cost effective and environment friendly as compared to conventional means of powering the desalination plant. The method used in achieving the aim of this research is categorized as pre-optimization stage, optimization stage, post-optimization stage (output and multi-criteria analysis) as shown in Fig. 1.

In the first stage, having identified the area under consideration, the information about the water requirements and consequently the electrical power demand of the desalination plant is carried out. Also, at this stage, the available energy resource, technical and cost data of system components, and constraints (technical, cost, environmental) of the proposed energy system are obtained. The input at this stage affects the overall operation and planning of the system, hence, it is important to carefully analyse this part of the framework. In estimating the capacities of components and stages of a desalination plant and its power requirements, various tools have been developed and used by researchers, some of which include ROSA, IMSDesign, Winflows, CSMPPro, TorayDS2,

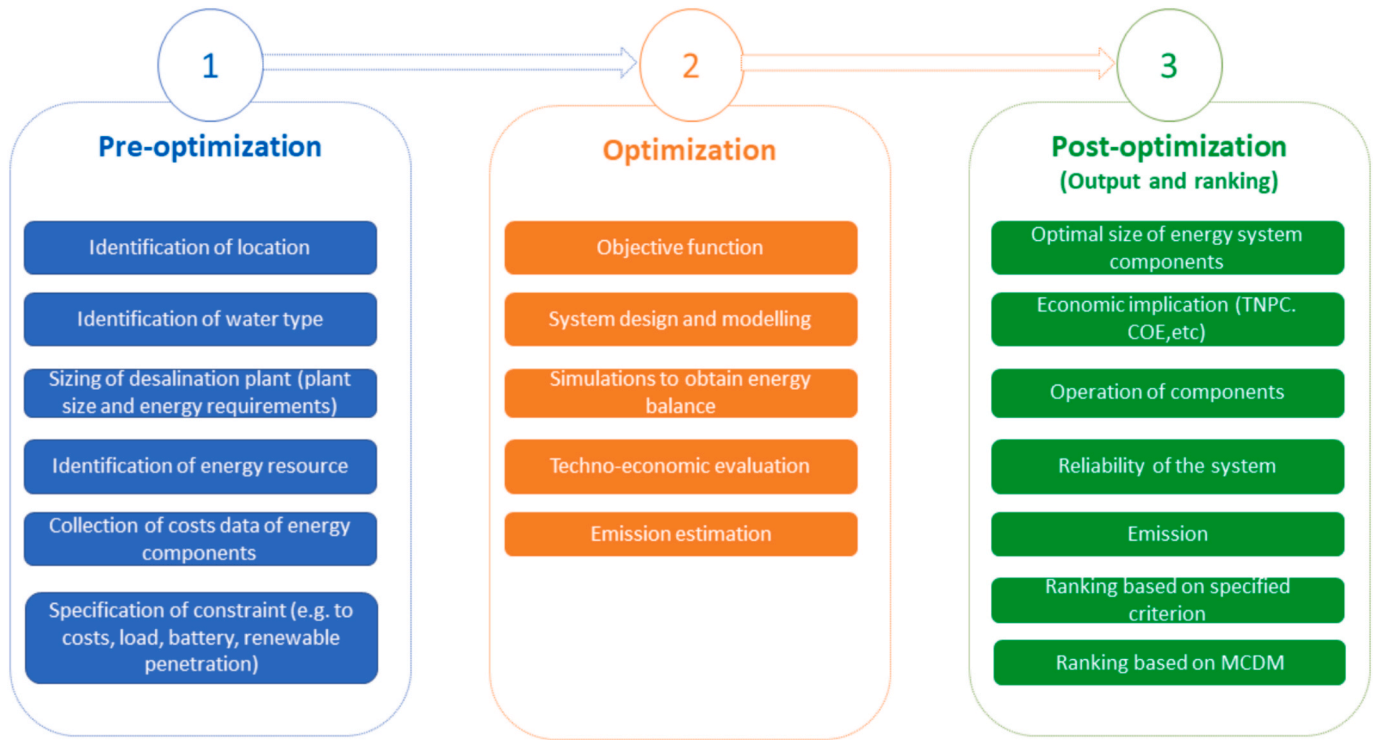


Fig. 1. Presented framework procedure.

KMSROPRO, TROI, and $Q+$ [28]. In the second stage, optimization process (Techno-economic) is done to determine the capacities of the various components of the technically feasible systems that can power the desalination plants. Also, the costs implication of these feasible energy systems over its lifetime is determined and used to rank the energy systems. The emissions attributed to the feasible energy systems are also estimated at this stage.

The last stage involves the extraction of the outputs, some of which include the sizes of the components, the life cycle cost, the cost of energy, excess energy, emissions and so on. The energy can be ranked based on any of these metrics. The reliability of the energy systems can also be obtained. The energy systems are ranked based on the total net present cost (TNPC). However, the system can also be ranked based on more than one of these metrics by using a multi-criteria decision method. Ranking based on multiple criteria increases the sustainability of the system[29].

Site description

South Africa over the years have faced consistent water scarcity and has resulted in water demand exceeding the available natural water (most especially those from river basins). This is especially because of intermittent rainfall patterns over the various climatic zones and very extreme evaporation rates. Apart from these, the country's water option is limited to harnessing surface water because the availability of ground of water is restricted by the geological structure of South Africa which is dominated by hard rocks. Furthermore, places where it is easy to access ground water, are plagued also with over-extraction, thereby precipitating water scarcity. Communities adjacent to places where major socio-economic activities such as industrial production and mining are usually also faced with the threats of water scarcity. Between mid-2017 and 2018, Cape Town almost ran out of water- one of the worst-ever drought-induced municipal water crises, residents became water-wise. Due to the perpetually growing population, positive economic growth forecasts, and prevailing efficiency levels, water demand in South Africa is anticipated to increase to 17.7 billion cubic meters by 2030.

Conversely, water supply is not expected to go beyond 15 billion cubic meters, signifying a 17 % (up 3.8 billion cubic meters) shortfall between water demand and supply. To explore a climate-independent means of providing the water deficit in South Africa, the hypothetical water desalination system is assumed to be in the coastal provinces of South Africa. There are four south African provinces that has border with the sea (Fig. 2), and they include KwaZulu-Natal (Durban: 29.8587° S, 31.0218° E), Eastern cape (Port Elizabeth: 33.9608° S, 25.6022° E), Western Cape (Cape Town), and Northern Cape (Port Nolloth: 29.2415° S, 16.9004° E).

Integration of hybrid power (wind-photovoltaic-diesel-battery) and seawater reverse osmosis systems for small-scale desalination applications.

Desalination plant description and load profile

The World Health Organization (WHO) recommend that an individual in a semi-dry climate would need approximately 20 L of fresh water daily[30], however, during Cape Town's 2017–2018 drought, the City capped household consumption at 100 L per person per day (Level 4), later reducing the cap to 87 L (Level 4B) and 50 L (Level 6B), before easing to 105 L in late 2018[31]. Based on this assumption, the proposed desalination plant which is designed to deliver 19kL of fresh water daily is expected to serve approximately 190 people per day. If the former assumption were considered, 950 people would benefit from the supply of fresh water. The flow diagram for the process of purification is shown in Fig. 3 with each number representing the streams. This desalination plant transforms seawater into fresh water through a systematic process. Seawater enters at the intake point (1) and flows to the pre-treatment facility (2), where debris and large particles are removed through screening and filtration. The treated water then passes through a high-pressure pump (triangular symbol) before entering the main processing stage (3). The core desalination unit uses reverse osmosis membranes to separate salt from water, producing two streams: clean drinking water that exits through node (7) for distribution, and concentrated brine that flows to treatment stage (4). The brine



Fig. 2. .
Source (<https://www.rilleyssolidarity.com/fr/introduction-to-the-provinces-2/>)

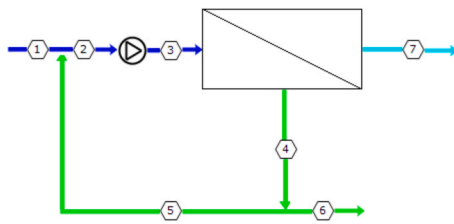


Fig. 3. Desalination plant process flow diagram showing seawater treatment through reverse osmosis system.

undergoes energy recovery processing (5) before final disposal (6). An energy-efficient recirculation loop connects the pre-treatment stage back to the brine processing system, maximizing operational efficiency while ensuring environmentally responsible waste management.

At the beginning of the process (stream 1), the TDS is assumed to be 35000 mg/L while the final output from the reverse osmosis (stream 7), the TDS is 505 mg/L.

Table 2 provides an overview of the essential chemical and power needs for the proposed desalination plant design obtained through the IMS design software. The pH of the feed water is 7.9, indicating its slightly basic nature. The desalination plant operates continuously at 19 m³/day capacity, producing 6,935 m³ of freshwater annually. This is obtained from a feed flow of 126.7 cubic meters per day, which leads to a reject flow of 107.7 cubic meters per day. Additionally, there is a recirculation flow of 0.2 cubic meters per day. The power calculations suggest that a pump/boost pressure of 38.4 bar is needed. The pump is expected to have an efficiency of 87 %, the motor an efficiency of 95 %, and the VFD an efficiency of 97 %. As a result, the total power requirement for pumping is estimated to be 7 kW, and the specific energy consumption for pumping is projected to be 8.87 kWh per cubic

Table 2

Overview of the essential chemical and power needs for the proposed desalination plant design.

Feature	Unit	Value
Train calculations		
Feed pH		7.9
Permeate recovery	%	15
Permeate flow	m ³ /day	19
Feed flow	m ³ /day	126.7
Reject flow	m ³ /day	107.7
Recirculation flow	m ³ /day	0.2
Power Calculations		
Pump/Boost pressure	Bar	38.4
Pump efficiency	%	87
Motor efficiency	%	95
VFD efficiency	%	97
Total pumping power	kW	7
Pumping specific energy	kWh/m ³	8.87

meter of permeate produced. Table 2. Flow and power requirements of the desalination plant.

Solar radiation

Fig. 4 displays the monthly solar radiation data for four locations that were examined in this study: Alexandra Bay, Jeffery Bay, Paternoster, and Ramsgate. All locations exhibit a similar seasonal pattern, with higher solar radiation values between December to February and lower values between May to July. Alexandra Bay and Paternoster exhibit greater daily solar radiation levels in comparison to Jeffery Bay and Ramsgate. The highest values are recorded in December for Alexandra Bay (7.684), Paternoster (7.766), and Jeffery Bay (6.589), but Ramsgate experiences its peak in January and February (5.254). The minimum

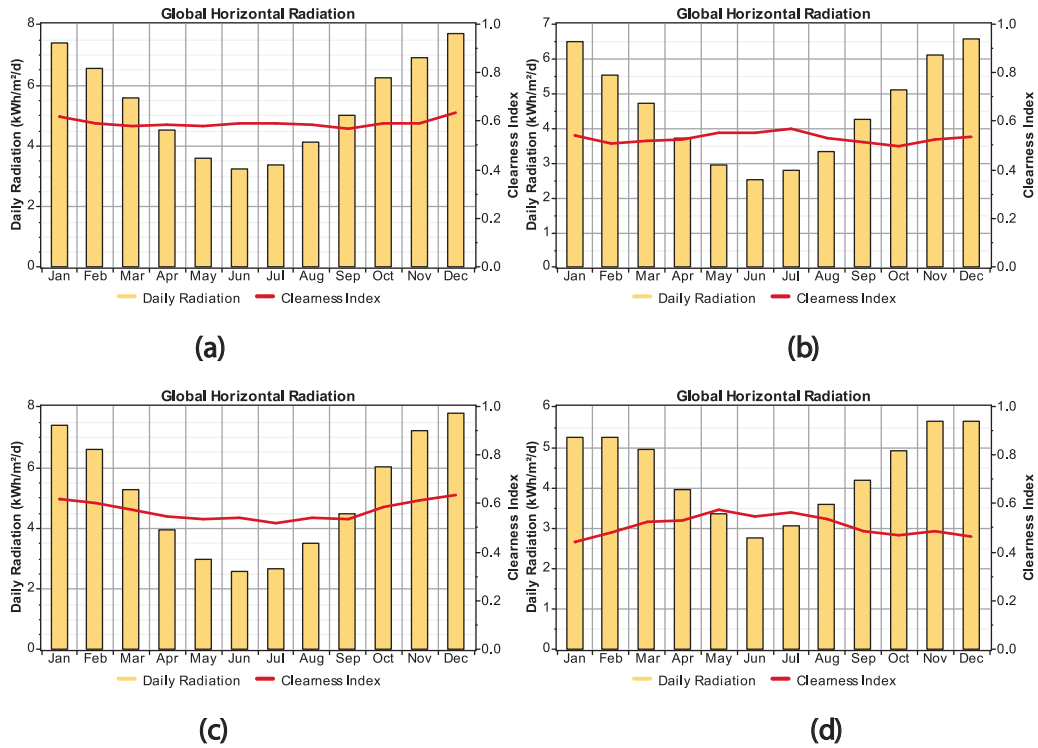


Fig. 4. Global horizontal radiation and clearness index for the locations (a) Alexandra Bay (b) Jeffery Bay (c) Paternoste (d) Ramsgate.

values are observed in June for Alexandra Bay (3.209), Jeffery Bay (2.506), and Paternoster (2.558), and in May for Ramsgate (3.352). Based on the statistics, Alexandra Bay and Paternoster appear to be better suited for solar energy applications that demand higher amounts of radiation, whereas Ramsgate consistently sees lower radiation levels throughout the year.

Description and size specification of system components

This integrated renewable energy system combines solar PV panels and battery storage to power a desalination plant (63kWh/d, 10 kW peak) (Fig. 5). Excess energy produces hydrogen via electrolysis, stored in tanks for later use in fuel cells. A converter manages AC/DC power flow between components, creating a sustainable water-energy nexus solution.

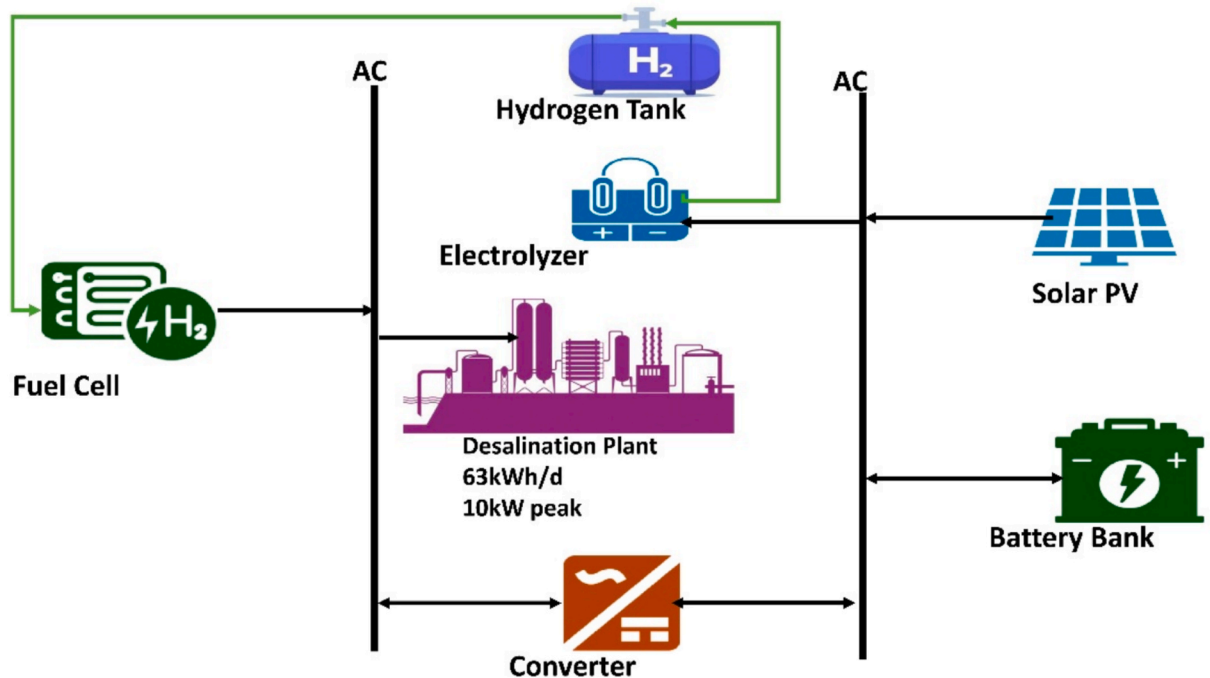


Fig. 5. Proposed power systems configuration.

PV panel array electricity production

Because of the rich solar resource in South Africa, solar PV (connected in the off-grid mode) is included as part of the energy system components that would support the desalination plant. The power output of a PV is typically based on single-diode model. For a PV array, the power output can be obtained by using equation (1).

$$P_{PVO} = P_r f_d \left(\frac{G_i}{G_{i-STC}} \right) [1 + (T_c - T_{c-STC}) \alpha_p] \quad (1)$$

where P_r denotes the rated output on the name plate of the PV panel at standard test conditions, f_d is the derating factor, G_i which is measured in W/m^2 is the incident solar radiation on the PV module, G_{i-STC} is the total irradiance measured under standard test conditions, α_p is the temperature coefficient, T_c is the temperature of the PV module, and T_{c-STC} is the temperature of the PV module under standard test conditions. In this study, it is assumed that a 250Wp monocrystalline PV panel with an efficiency of 16.7 % at standard test conditions would cost \$550 at the beginning of the project. Other relevant details about the PV panel include replacement cost-\$550, O&M cost-\$10/year, lifetime-20 years, derating factor-80 %, nominal operating temperature- 47 °C and temperature coefficient of power (-0.5 %/°C)

Energy storage systems

The proposed energy system is designed to uses both a battery storage system and a hydrogen storage system as backup. The fuel cell, electrolyzer, and hydrogen storage tank make up the hydrogen energy system while a Sealed Rechargeable Deep Cycle battery is selected for the system.

Electrolyzer

The electrolyzer is essential for transforming electrical power generated by photovoltaic (PV) panels into hydrogen gas using the process of water electrolysis. The Proton Exchange Membrane (PEM) electrolyzer was selected from several electrolysis technologies due to its exceptional efficiency, extended operational longevity, and capacity to accommodate the fluctuating power outputs typically found in Renewable Energy Sources (RES). The efficiency of the PEM electrolyzer and the electrical energy consumption for hydrogen production are crucial elements in the scope of this study. Equations (12) to (14) provides the mathematical expression for hydrogen generated by the PEM electrolyzer.

$$\alpha \dot{H}_2 = \eta_f \left(\frac{N_s \times I_e}{n \times F} \right) \quad (2)$$

$$P_{com} = \alpha \dot{H}_2 \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} \right] \left(\frac{\gamma}{\gamma-1} \right) \left(\frac{T}{\eta_{com}} \right) R \quad (3)$$

$$FC_{ele} = LHV_h \times \alpha \dot{H}_2 \times \rho_h \times \eta_{fc} \quad (4)$$

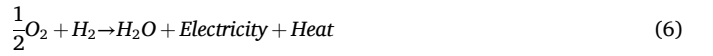
The P_2 and P_1 represent the inlet and outlet pressure, respectively, R stands for the gas constant, γ represents the polytropic coefficient, η_{comm} denotes the efficiency of the compressor, $\alpha \dot{H}_2$ refers to the mass flow rate of hydrogen, ρ_h represents the density of compressed $\dot{H}_2$, LHV_h signifies the lower heating value of hydrogen gas, and η_{fc} represents the fuel cell efficiency.

Fuel cell modelling

A fuel cell system is a device that uses redox reactions to directly transform the chemical energy of a fuel and an oxidizing agent into electricity. Because it bypasses the intermediate phases of converting chemical energy to thermal energy and subsequently to mechanical

energy, this approach is more efficient than traditional combustion-based systems. Because the byproducts are typically water and heat, fuel cell devices are also environmentally friendly. Fuel cell systems are classified according to the type of electrolyte used, with each type having its own set of advantages, limitations, and applications. Depending on the electrolyte used, fuel cell systems can be divided into different categories, such as proton exchange membrane fuel cells (PEMFCs), solid oxide fuel cells (SOFCs), molten carbonate fuel cells (MCFCs), alkaline fuel cells (AFCs), and phosphoric acid fuel cells (PAFCs).

The electrochemical principles of redox reactions and ion transport through an electrolyte serve as the foundation for the theory underlying fuel cell electricity generation. Surplus electricity generated by the photovoltaic system can be utilized to power an electrolyzer, which produces hydrogen gas; this hydrogen gas is then stored in a hydrogen storage tank. Hydrogen functions as a storage to be utilized during periods of inadequate availability of alternative energy sources. The electrolyzer employs electrical current to decompose water via the process of electrolysis as follows [32]:



Battery bank modelling

Just like the diesel generator, the battery energy storage is also a form of backup for the proposed energy system. In this study it is assumed that the battery is charged when there is excess energy and discharged when there is insufficiency of energy from the renewable energy sources. A 6FM200D-X battery with a nominal voltage of 12 V, 200Ah (2.4kWh nominal capacity) and lifetime throughput of 917kWh is selected. Lead-lead dioxide systems are used in rechargeable batteries. The battery autonomy is calculated as [32]:

$$Batt_{aut} = \frac{n_b V_n Q_n \left(\frac{100 - SOS_{min}}{100} \right) (24hr/day)}{D_{pry,ave} (1000Wh/kWh)} \quad (7)$$

where $D_{pry,ave}$ is average primary energy demand, SOS_{min} is minimum state of charge (%), Q_n is nominal capacity of a single battery (Ah), V_n is the nominal voltage of a single battery (V), and n_b is the number of battery.

This study utilizes differential equations to calculate the upper limit of power that the battery can deliver or receive at any given time interval Δt . This is determined using Equations (8) and (9) [32].

$$P_{batt,dmax,kbm} = \frac{QKC(1 - e^{-k\Delta t}) + KCQ_1 e^{-k\Delta t}}{C(K\Delta t - 1 + e^{-k\Delta t}) + 1 - e^{-k\Delta t}} \quad (8)$$

$$P_{batt,dmax,kbm} = \frac{QKC(1 - e^{-k\Delta t}) - KCQ_{max} + KCQ_1 e^{-k\Delta t}}{C(K\Delta t - 1 + e^{-k\Delta t}) + 1 - e^{-k\Delta t}} \quad (9)$$

Q represent the total energy in the battery at the beginning of the time step, Q_1 represent the available energy at the start of the time step, and Δt represent the length of the time step. Q_{max} is the maximum amount of energy that may be stored in the battery, K is the rate constant, and C is the capacity ratio. The battery's lifetime throughput is given as:

$$Q_{lifetime} = f_c \times DoD \left(\frac{V_n Q_{max}}{1000W/kW} \right) \quad (10)$$

Economic details of components

Table 3 summarizes the components of an energy system, including their size, capital cost, replacement cost, and operation and

Table 3

Techno-economic description of system components including their size, capital cost, replacement cost, and operation and maintenance.

Component	Size	Capital (\$)	Replacement (\$)	O&M Cost
Fuel Cell	3,000 kW	3,000	3,000	\$0.01/hr
Battery: Vision 6FM200D	Quantity: 1	357	350	\$1.5/yr
Converter	1 kW	37	36	\$2/yr
DC Electrolyzer	8 kW	2,700	2,700	\$3/yr
Hydrogen Tank	8 kg	3,100	3,100	\$4/yr

maintenance (O&M) costs. The system is made up of a 3,000 kW fuel cell with a capital and replacement cost of \$3,000 per and an operating and maintenance cost of \$0.01 per hour. The Vision 6FM200D battery has an initial cost of \$357, a cost of \$350 for replacement, and an annual operating and maintenance cost of \$1.5. The converter has a power rating of 1 kW (kW), an initial cost of \$37, a cost of \$36 for replacement, and an annual operation and maintenance (O&M) cost of \$2. The DC electrolyzer, which has a power capacity of 8 kW, has a capital cost and replacement cost of \$2,700 each, and an operation and maintenance cost of \$3 per year. Finally, the hydrogen tank, with a capacity of 8 kg of hydrogen, has a capital and replacement cost of \$3,100 per, and an O&M cost of \$4 per year. The fuel cell and hydrogen tank have the highest capital and replacement costs, while the O&M costs vary across components, with the fuel cell having the highest hourly O&M cost and the hydrogen tank having the highest annual O&M cost.

Total net present cost (TNPC)

The total net present cost of a system is determined by subtracting the present value of all the revenue it generates throughout its lifespan from the present value of all the costs it accrues throughout its lifespan. The costs include capital expenditures, replacement expenses, operational and maintenance costs, fuel costs, pollution penalties, and costs related to obtaining power from the grid. Revenues comprise the monetary worth acquired from salvage operations and the income received from sales made through the grid. The TNPC is calculated using Equation (11), [33].

$$TNPC = \frac{TC_{ann}}{F_{cr}(i, X_{proj})} \quad (11)$$

were, TC_{ann} is the achieved total annual cost, while F_{cr} is the capital recovery factor. i is the interest rate in % and X_{proj} represent the project lifespan. This work adopted i as 12 % and X_{proj} to be 25 years.

Levelised cost of energy (LCOE)

Levelised cost of energy (COE) is the average cost per kWh of useful electrical energy generated by the system. COE is calculated by dividing the yearly cost of producing electricity (the total annualized cost less the cost of feeding the thermal load) by the total useful electric energy production. The equation for determining COE is given by Equation (12) [33]:

$$LCOE = \frac{TC_{ann} - C_{boiler}E_{thermal}}{E_{pry,AC}} \quad (12)$$

Where $E_{pry,AC}$ is the total amount of electricity demand supplied by the source of electricity generation. The second term in the numerator represents the portion of the yearly cost that comes from serving the thermal load. In the systems presented in this study that do not provide a thermal load ($E_{thermal}$), this term is zero.

System reliability

To evaluate the reliability of each hybrid energy system design, the study considered the Primary Load Ratio (PLR). The PLR evaluates the ability of each of the systems to meet the combined load requirements of both essential and flexible components. The PLR was estimated as the ratio between the AC primary load served and the overall energy demand including both the AC primary load (essential desalination operations) and the load supplied to the electrolyzer (flexible hydrogen production). This metric represents system-wide energy delivery performance rather than essential load reliability specifically:

$$PLR(\%) = \frac{ACPrimaryLoadServed}{ACprimaryloadserved + electrolyzerloadserved} \times 100 \quad (13)$$

The Energy Not Supplied (ENS) was used to measure the annual total energy deficit, which is the difference between the AC load served and the total load demand as expressed by:

$$ENS \left(\frac{kWh}{yr} \right) = Totalloaddemand - ACprimaryloadserved \quad (14)$$

The Loss of Power Supply Probability (LPSP), a reliability risk indicator, was calculated as the fraction of total demand that was not met:

$$LPSP = \frac{ENS}{Totalloaddemand} = 1 - Reliability \quad (15)$$

These indicators provide a holistic measure of the system's performance under both load-following (LF) and cycle-charging (CC) dispatch strategies.

Multi-criteria decision making

A two-stage Multi-Criteria Decision-Making (MCDM) framework was employed. In Stage 1, the Entropy Method was used to determine the objective weights of the decision criteria. In Stage 2, the weighted decision matrix was analysed using WSM, WASPAS, EDAS, and ARAS to rank the alternatives. The selection of the Entropy-WSM-WASPAS-EDAS-ARAS framework was motivated by the need to combine objective weighting with robust ranking across multiple decision contexts. The Entropy method was chosen to determine criterion weights objectively, eliminating subjective bias by quantifying the degree of information variability within the dataset [34]. WSM was selected for its simplicity and effectiveness in additive aggregation [35], while WASPAS was included to integrate both additive and multiplicative decision logics, thereby enhancing ranking stability [36]. EDAS was chosen for its ability to evaluate alternatives relative to an average solution, providing a balanced view of both positive and negative deviations, which is valuable in performance benchmarking [34]. ARAS was incorporated for its straightforward ratio-based approach, which normalizes performance against the optimal alternative and is well-suited for both cost and benefit criteria [34]. This combination ensures a comprehensive analysis, cross-validates results through methodological diversity, and strengthens the reliability of final rankings.

Entropy weighting

The decision matrix $X = [x_{ij}]$ is normalized by means of um-based proportional normalization [34]:

$$n_{ij} = \frac{x_{ij}}{\sum_{i=1}^m x_{ij}} \quad (16)$$

The entropy for each for each criterion j was computed as

$$E_j = -k \sum_{i=1}^m P_{ij} \ln(P_{ij}), k = \frac{1}{\ln(m)} \quad (17)$$

The degree of diversification d_j and the weight w_j were calculated as:

$$d_j = 1 - E_j, w_j = \frac{d_j}{\sum_{i=1}^n d_j} \quad (18)$$

Weighted sum model (WSM)

The WSM score for alternative i is [35]:

$$S_i = \sum_{j=1}^n w_j \times r_{ij} \quad (19)$$

where r_{ij} is the normalized value of criterion j for alternative i .

Weighted Aggregated sum product assessment (WASPAS)

WASPAS combines WSM and the Weighted Product Model (WPM) [36]:

$$Q_i^{(1)} = \sum_{j=1}^n w_j \times r_{ij} \quad (20)$$

$$Q_i^{(2)} = \prod_{j=1}^n (r_{ij})^{w_j} \quad (21)$$

$$Q_i = \lambda Q_i^{(1)} + (1 - \lambda) Q_i^{(2)}, \lambda = 0.5 \quad (22)$$

Evaluation based on distance from average solution (EDAS)

The average performance for each criterion is [34]:

$$\bar{x}_j = \frac{\sum_{i=1}^m x_{ij}}{m} \quad (23)$$

Positive Distance from Average (PDA) and Negative Distance from Average (NDA) are calculated as:

$$PDA_{ij} = \begin{cases} \frac{x_{ij} - \bar{x}_{ij}}{\bar{x}_{ij}}, & \text{if benefit criterion} \\ \frac{\bar{x}_{ij} - x_{ij}}{\bar{x}_{ij}}, & \text{if cost criterion} \end{cases} \quad (24)$$

$$NDA_{ij} = \begin{cases} \frac{\bar{x}_{ij} - x_{ij}}{\bar{x}_{ij}}, & \text{if benefit criterion} \\ \frac{x_{ij} - \bar{x}_{ij}}{\bar{x}_{ij}}, & \text{if cost criterion} \end{cases} \quad (25)$$

The final score is given by:

$$AS_i = \frac{1}{2} (NPDA_i + (1 - NNDA_i)) \quad (26)$$

Additive ratio assessment (ARAS)

The normalized performance values are [34]:

$$r_{ij} = \begin{cases} \frac{x_{ij}}{\max_i x_{ij}}, & \text{if benefit criterion} \\ \frac{\min_i x_{ij}}{x_{ij}}, & \text{if cost criterion} \end{cases} \quad (27)$$

The weighted sum S_i is computed, and the utility degree K_i is

$$S_i = \sum_{j=1}^n w_j \times r_{ij} \quad (28)$$

$$K_i = \frac{S_i}{S_o} \quad (29)$$

where S_o is the optimal score.

Results and discussions

This section presents the results of system sizing, economic parameters, and operational characteristics for the optimal energy systems for four locations (Alex, Jeff, Pat, Ram) (Table 4). Table 5 presents comparative economic performance metrics for hybrid renewable energy systems across the four coastal locations when benchmarked against diesel-only generation. The specific energy consumption (SEC) of the proposed renewable-powered desalination systems ranged between 3.27 and 3.29 kWh m⁻³, indicating a narrow variation among the four locations. These values are within the typical range reported for reverse-osmosis (RO) desalination plants powered by renewable energy sources, generally between 3 and 6 kWh m⁻³, confirming the system's energy efficiency and technical feasibility. From Table 1, other studies show desalination plants with energy demands such as 4.38 kWh m⁻³ [7]; Ibrahim 2020 – PV/Wind/Diesel hybrids), 4 kWh m⁻³ [15] – Iran PV/WT/HS systems), and 0.83 kWh m⁻³ (Mostafaeipour 2019 – Iran PV/Battery system). When compared with these benchmarks, the obtained SEC values are lower than the 4–4.38 kWh m⁻³ reported for hybrid fossil-assisted systems and closer to the optimal range for medium-scale RO configurations, demonstrating improved performance through solar-hydrogen integration. Correspondingly, the Levelized Cost of Water (LCOW) varied from 0.796 to 0.896 USD m⁻³, with the lowest value observed at the site with the highest solar resource potential. The relatively stable SEC but variable LCOW suggests that economic performance is more sensitive to local resource availability and system configuration than to the desalination process itself. These results demonstrate that integrating solar PV, fuel-cell, and hydrogen subsystems can maintain consistent energy demand for desalination while reducing overall water production costs compared with conventional fossil-powered systems. The detailed analysis for each location is presented in subsequent subsections. Although the present analysis focuses on renewable-powered desalination and hydrogen generation, the concept can be further extended to include re-electrification opportunities from salinity-gradient power. Techniques such as pressure-retarded osmosis or reverse electrodialysis could recover a portion of the chemical potential from brine streams, converting salinity differences into usable electricity. Integrating such energy-recovery modules into future systems would close the Power-to-Hydrogen-to-Water loop, improve overall energy circularity, and reinforce the sustainability of renewable-driven desalination.

Alexandra Bay

The optimal energy system architecture for Alexandra Bay (Alex) consists of a 20 kW PV array, 30 kW fuel cell, 30 kW/72 kWh lithium-ion battery, 10 kW inverter, 7.5 kW rectifier, 10 kW electrolyzer, and 80 kg hydrogen tank. This hybrid system utilizes renewable solar energy as the primary source, with the battery and fuel cell providing supplementary power as needed. The PV array accounts for the majority (97 %) of the annual electrical production at 34,406 kWh/yr, with a capacity factor of 19.6 %. Excess PV generation charges the batteries and powers the electrolyzer to produce hydrogen. The fuel cell for this energy system operates at a 0.379 % capacity factor as a secondary generator, while the battery storage provides 16.6 h of autonomy at a 43.2 kWh usable capacity over a 10-year lifetime. The hydrogen tank stores the electrolytically produced hydrogen to power the fuel cell with a total annual hydrogen production of 132 kg, of which 59.8 kg is consumed by the fuel cell. This provides a hydrogen tank autonomy of 1022 h. The optimal energy system achieves 100 % renewable penetration with no unmet AC

Table 4
Optimal system configurations and performance metrics for hybrid renewable energy systems across the 4 locations.

Location	PV (kW)	FC (kW)	Battery (No.)	Converter (kW)	Electrolyzer (kW)	H ₂ Tank (kg)	Dispatch Strategy	Total NPC (\$)	Total O&M Cost (\$/yr)	Total Ann. Cost (\$/yr)	COE (\$/kWh)	PV Production (kWh/yr)	FC Production (kWh/yr)	AC Primary Load Served (kWh/yr)	Electrolyzer Load Served (kWh/yr)	Excess Electricity (kWh/yr)	NOx Emissions (kg/yr)	Battery Throughput (kWh/yr)	Battery Life (yr)	Specific Energy Consumption (kWh/m ³)	Levelized Cost of Water (\$/m ³)
Alex	20	30	30	10	10	80	LF	122,742	171	5518	0.242	34,406	997	22,849	6116	3399	3	2729	10	3.29	0.796
Jeff	20	30	30	10	10	80	LF	136,000	217	6,114	0.27	30,143	1,824	22,653	6,232	25	6	3,392	8.1	3.27	0.882
Pat	20	30	30	10	10	80	CC	128,515	183	5,778	0.255	32,704	2,210	22,662	9,025	40	8	3,201	8.6	3.27	0.833
Ram	25	30	30	10	10	80	LF	138,239	200	6,215	0.272	36,479	1,554	22,849	7,607	4,537	5	3,144	8.8	3.29	0.896

Table 5

Economic Performance Metrics of Hybrid Renewable Energy Systems Compared to Diesel-Only Generation.

Metric	Alex	Jef	Pat	Ram
Present worth (\$)	90,835	77,577	85,062	75,338
Annual worth (\$/yr)	4,084	3,488	3,824	3,387
Return on investment (%)	7.31	6.91	7.14	6.77
Internal rate of return (%)	4.12	3.74	3.94	3.55
Simple payback (yr)	18.4	22	21.9	23.3
Discounted payback (yr)	23.7	23.9	23.8	24.4

load. The hydrogen tank storage level chart for Alexandra Bay displays hourly hydrogen availability throughout the year (Fig. 6). January begins with lower storage levels (16–32 kg) which steadily increase through February and March (reaching 48 kg). From April through December, the tank maintains near-maximum capacity (64 kg), with minor depletions during July and August. Furthermore, the hybrid configuration results in zero net carbon dioxide emissions annually. The levelized cost of energy (COE) is estimated to be \$0.242/kWh over the project lifetime, while the total net present cost (TNPC) of the energy system is estimated at \$122,742. Fig. S1 gives the cost contributions of each of the system components for a 25-year period; the battery is replaced in years 10, 20 and 25, while the electrolyzer is replaced in year 14. As for the solar PV panel, replacements are made in years 20 and 25. The annualized cash flow (Fig. S2) indicates major costs from capital (\$7,151) and replacement (\$1,852.8), with minor operating expenses (\$171.15) with no fuel cost while the salvage recovery is \$3,656.5). When compared to diesel-only generation, Alexandra Bay demonstrates superior economic viability (Table 5) with a present worth of \$90,835, annual worth of \$4,084/yr, return on investment of 7.31 %, internal rate of return of 4.12 %, and the shortest simple payback period of 18.4 years, indicating the highest financial returns from integrating renewable energy and hydrogen storage.

The emissions attributed to carbon monoxide, unburned hydrocarbons, and particulate matter are negligible, while 3.47 kg of nitrogen oxide is emitted annually as the primary pollutant.

Examining the power generation diagram for a typical week in June at Alexandra Bay provides valuable insights into the operational dynamics of the hybrid renewable energy system powering the desalination plant. Fig. 7 illustrates the interplay between the three key components of the Alexandra Bay system. Figure 7 confirms that solar PV serves as the dominant power source at Alexandra Bay, with peak generation reaching 15–16 kW during optimal daylight hours. This aligns with the study's finding that the PV array accounts for the majority (97 %) of annual electrical production at 34,406 kWh/yr, with a capacity factor of 19.6 %. The distinct daily solar cycles show strong but intermittent generation, with approximately 7–8 h of productive output each day followed by zero production during nighttime. This pattern is consistent with Alexandra Bay's solar resource profile, which the study identified as one of the strongest among the examined locations, with global horizontal radiation of up to 7.684 kWh/m²/day in peak months. Notably, during June (a winter month in South Africa when solar radiation is at its annual minimum), the PV system still generates sufficient power to handle most of the load requirements.

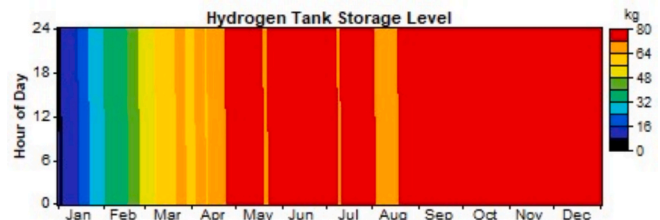


Fig. 6. Annual hourly hydrogen availability for the energy system for Alex.

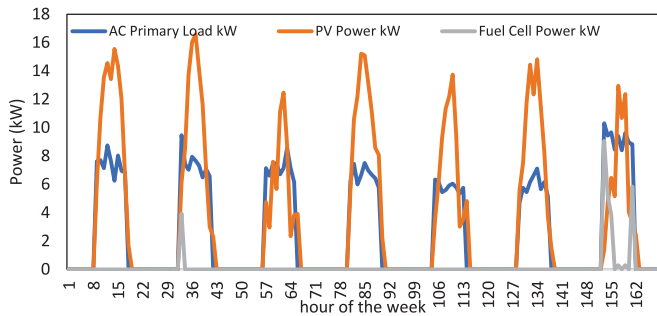


Fig. 7. Operational dynamics of the hybrid renewable energy system in Alex.

Furthermore, the operational dynamics shows the fuel cell activating primarily during evening hours when solar generation drops, with occasional daytime activation during periods of lower solar output or higher demand. These brief operational periods (reaching 8–10 kW) align perfectly with the study's finding that the fuel cell operates at a 0.379 % capacity factor as a secondary generator. The fuel cell's pattern of operation demonstrates its critical role in maintaining system reliability despite its limited operational hours. The operational dynamics show that fuel cell activation is precisely targeted to periods of energy deficit, highlighting the effectiveness of the system's dispatch strategy, which the study identified as "Load Following" (LF) for Alexandra Bay.

The Alexandra Bay sensitivity plot (Fig. S3) evaluates optimal system configurations across FC Capital Multiplier and FC Replacement Multiplier ranges from 0.0 to 1.0. Two system architectures compete: PV/FC/Gen/Electrolyzer/Battery and PV/Gen/Battery. A diagonal transition boundary separates these configurations. The hydrogen-integrated PV/FC/Gen/Electrolyzer/Battery system becomes optimal only when both FC capital and replacement multipliers stay below approximately 0.20 to 0.30, occupying roughly 15 to 20 percent of the parameter space. Beyond this threshold, the PV/Gen/Battery configuration dominates, covering 80 to 85 percent of all scenarios. The diagonal boundary shows that capital and replacement costs influence system selection proportionally, requiring simultaneous reductions in both parameters for hydrogen technology adoption. When either multiplier exceeds 0.30, the battery-based system proves more economical regardless of the other parameter's value. These findings indicate that fuel cell viability in Alexandra Bay requires cost reductions approaching 70 to 80 percent below baseline values for both initial investment and lifecycle replacement expenses, highlighting economic barriers to hydrogen-based renewable energy deployment under current or moderately improved pricing conditions.

Jeffery Bay

The optimal system configuration for Jeffery Bay (Jef) is like that of Alexandra Bay, with a 20 kW PV array, 30 kW fuel cell, 30 kW/72 kWh lithium-ion battery, 10 kW inverter, 7.5 kW rectifier, 10 kW electrolyzer, and 80 kg hydrogen tank. The levelized cost of energy is slightly higher at \$0.270/kWh. The PV array contributes 94 % of the annual electrical production at 30,143 kWh/yr, with a 17.2 % capacity factor. The fuel cell operates at a 0.694 % capacity factor to provide 1,824 kWh/yr. There is an energy deficit, with 196 kWh/yr of unmet AC load and 228 kWh/yr of capacity shortage. The battery capacity provides 16.6 h of autonomy, but experiences greater throughput (3,392 kWh/yr), reducing expected life to 8.11 years. Hydrogen production matches closely to consumption by the fuel cell at 134 kg/yr produced and 109 kg/yr consumed. The hydrogen tank maintains over 1000 h of autonomy. The monthly hydrogen tank storage level for the optimal energy system shows low levels of hydrogen for majority of the month of the year except for March which has moderate hydrogen storage (Fig. 8). With a reduced PV output, the fuel cell at this location is utilized more as compared to the one attached to the system at Alexandra Bay, nearly

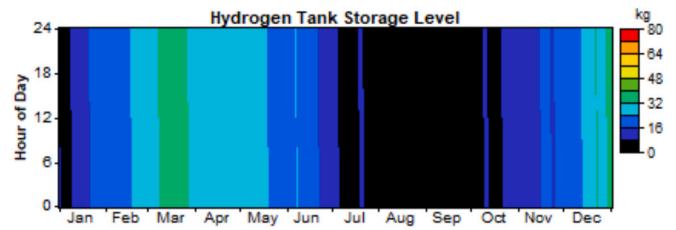


Fig. 8. Annual hourly hydrogen availability for the energy system for Jeffery Bay.

doubling the operating hours of the fuel cell in Alexandra Bay.

Fig. 9 shows the operational dynamics of the hybrid renewable energy system in Jeffery Bay. The PV array (reaching peaks of 16 kW) demonstrates strong but intermittent generation during daylight hours, with notable fluctuations indicating variable weather conditions. Unlike Alexandra Bay, Jeffery Bay shows more frequent and higher-amplitude fuel cell activation (up to 10 kW), particularly around hours 28–32, 45–50, and 150–165. This increased fuel cell utilization aligns with the study's finding that Jeffery Bay's system has a fuel cell capacity factor of 0.694 % (nearly double Alexandra Bay's 0.379 %), producing 1,824 kWh/yr. The load profile remains relatively consistent at 6–8 kW with occasional 10 kW peaks. This operational pattern explains Jeffery Bay's higher levelized energy cost (\$0.270/kWh vs Alexandra Bay's \$0.242/kWh) and increased nitrogen oxide emissions (6.35 kg/yr), resulting from greater dependence on the fuel cell during periods of insufficient solar generation.

Fig. S4 shows the annualized cost summary for Jeffery; the total project cost of \$236,999.7 is largely driven by capital costs, which account for 67.1 % of total expenditure, followed by replacement costs at 21.1 % and operating expenses at 2.0 %. The salvage value, representing 32.9 % recovery, significantly offsets earlier expenses, reflecting efficient cost management and long-term financial sustainability. Replacements occurred for the battery at 9th and 17th year, converter at 10th year, electrolyzer at 14th and 25th year, and PV module in years 20 and 25 (Fig. S5).

Relative to diesel-only systems, Jeffery Bay shows moderate improvement with a present worth of \$77,577, annual worth of \$3,488/yr, return on investment of 6.91 %, internal rate of return of 3.74 %, and a simple payback period of 22 years (Table 5), reflecting positive but less pronounced economic benefits from hybrid renewable integration.

The Jeffery Bay sensitivity analysis (Fig. S6) shows PV/FC/Gen/Electrolyzer/Battery and PV/Gen2/Battery dominate the results. A diagonal transition boundary separates the parameter space, with the hydrogen-integrated system achieving economic optimality only when FC capital costs remain below approximately 0.20 to 0.40 and replacement costs stay below 0.30 to 0.50. This represents roughly 20 to 25 percent of the examined scenarios. The PV/Gen/Battery configuration proves superior across 75 to 80 percent of the parameter space,

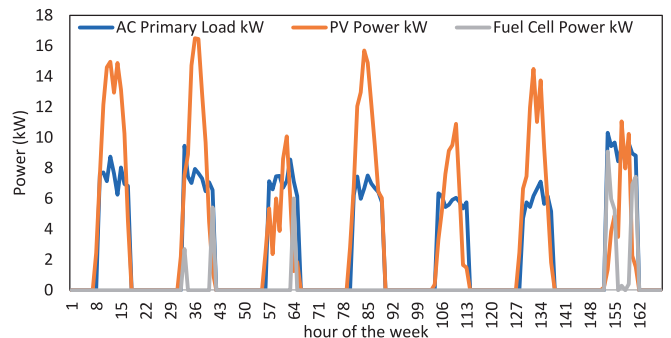


Fig. 9. Operational dynamics of the hybrid renewable energy system in Jeffery Bay.

particularly when either cost multiplier exceeds 0.40. The nearly linear boundary suggests proportional sensitivity to both capital and replacement costs, requiring simultaneous reductions for hydrogen system viability. When capital multipliers rise above 0.40 or replacement multipliers exceed 0.50, the battery-diesel system consistently delivers better economics. These results indicate that fuel cell deployment at Jeffery Bay demands cost reductions of 60 to 80 percent below baseline values, with both initial investment and lifecycle maintenance requiring comparable improvements to enable competitive hydrogen-based energy storage solutions.

Paternoste

The best system configuration for Paternoste consists of a solar photovoltaic (PV) array with a capacity of 20 kW, a fuel cell with a capacity of 30 kW, a lithium-ion battery bank with a storage capacity of 30 kWh, an inverter with a capacity of 10 kW, a rectifier with a capacity of 7.5 kW, an electrolyzer with a capacity of 10 kW, and an 80 kg hydrogen tank. The TNPC over a 25-year period is \$128,515, resulting in a levelized cost of \$0.255/kWh. The PV system contributes the most to capital costs (\$29,078), followed by the fuel cell system (\$31,116). Battery replacement is likewise expensive, costing \$18,676. The photovoltaic (PV) array has an annual energy generation of 32,704 kWh, which accounts for 94 % of the overall renewable energy production. The solar production profile has a strong alignment with the load shape, as evidenced by its capacity factor of 18.7 %. The fuel cell's contribution to the overall generation is limited to 2,210 kWh each year, which accounts for just 6 % of the total generation. The fuel cell operates at a capacity factor of 0.841 %. There is a significant unmet AC demand of 187 kWh per year, resulting in a capacity deficiency of 219 kWh per year. This suggests that there is insufficient capacity during peak periods. Fig. 10 shows that the hydrogen tank storage level peaks between March and June, indicating surplus production, but drops sharply in August and September due to low generation or high consumption.

Fig. 11 depicts a typical week in June for the Paternoste system, showcasing the interplay between power sources and demand. The PV array demonstrates strong but variable generation, with peaks reaching 15–16 kW during optimal daylight conditions. Most notably, Paternoste exhibits the most significant fuel cell utilization among the studied sites, with substantial activation around hours 60–70 (reaching nearly 14 kW) and again around hours 150–165 (exceeding 10 kW). This fuel cell operation aligns with the study's finding that Paternoste has the highest fuel cell capacity factor at 0.841 %, producing 2,210 kWh annually. The AC Primary Load maintains a relatively consistent profile of 6–8 kW with occasional peaks. Unlike the other locations, Paternoste employs a “Cycle Charging” (CC) dispatch strategy rather than “Load Following” (LF), which explains the more aggressive fuel cell operation pattern. This operational characteristic contributes to Paternoste's higher electrolyzer utilization (9,025 kWh/yr) and explains its intermediate levelized cost of energy at \$0.255/kWh, positioned between Alexandra Bay's \$0.242/kWh and Jeffery Bay's \$0.270/kWh. The total project cost of \$228,655 (Fig. S7) is primarily driven by capital costs (69.6 %), followed by replacement (18.2 %) and operating expenses (1.8 %), while

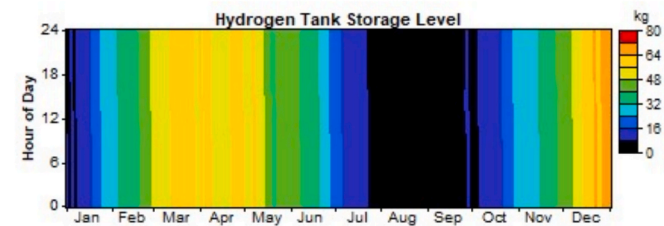


Fig. 10. Annual hourly hydrogen availability for the energy system for Paternoste.

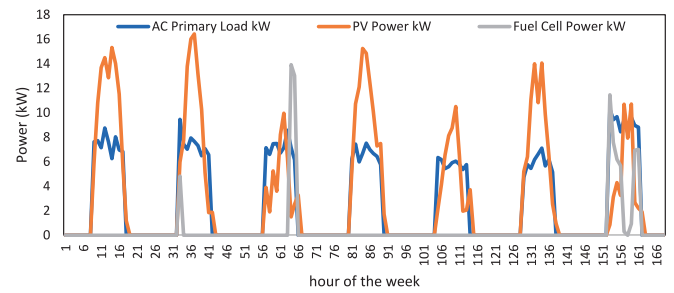


Fig. 11. Operational dynamics of the hybrid renewable energy system in Paternoste.

the salvage recovery (−\$76,170) offsets a substantial portion of prior expenditures as shown in Fig. S8. Mid-cycle replacements occur for the battery, converter, and electrolyzer, ensuring sustained performance.

Compared to conventional diesel generation, Paternoster exhibits intermediate economic advantages with a present worth of \$85,062, annual worth of \$3,824/yr, return on investment of 7.14 %, internal rate of return of 3.94 %, and a simple payback period of 21.9 years, demonstrating substantial cost savings from renewable-hydrogen hybridization (Table 5).

The Paternoster sensitivity plot (Fig. S9) also shows a diagonal transition boundary that divides the parameter space, with the hydrogen-integrated PV/FC/Gen/Electrolyzer/Battery system optimal only when both FC capital and replacement multipliers remain below approximately 0.20 to 0.30. This favorable region occupies roughly 15 to 20 percent of the total parameter space. The PV/Gen/Battery configuration dominates the remaining 80 to 85 percent of scenarios, becoming economically superior when either cost multiplier exceeds 0.30. The diagonal boundary pattern indicates that capital and replacement costs exert proportional influence on system selection, requiring concurrent reductions in both parameters for hydrogen technology viability. These results suggest that fuel cell deployment at Paternoster requires cost reductions approaching 70 to 80 percent below baseline values for both initial capital and lifecycle replacement expenses. The findings highlight substantial economic barriers facing hydrogen-based systems, indicating that conventional battery storage with diesel backup remains more cost-effective under current or moderately improved fuel cell pricing scenarios.

Ramsgate

The hybrid system architecture at Ramsgate (Ram) utilizes a 20 kW PV array as the primary generation source, supplemented by a 30 kW fuel cell, 30 kWh lithium-ion battery, 10 kW electrolyzer, and 80 kg hydrogen tank for energy storage and flexible dispatchable power. This integrated configuration combines the benefits of solar PV with energy storage technologies to enable high renewable penetration. The PV array supplies 34,406 kWh annually, meeting 97 % of the 22,849 kWh load. The minimal 3 % gap is filled by the fuel cell producing 997 kWh yearly. The battery's 2,441 kWh throughput shifts solar power to meet evening loads, while the electrolyzer leverages surplus PV to produce 132 kg of hydrogen for long-term storage. This coordinated operation results in zero unmet AC load and a 100 % renewable fraction, despite solar intermittency. Over the project lifetime, the system has a total net present cost of \$122,742, with solar PV incurring the largest capital costs. The levelized cost of \$0.242/kWh demonstrates economic viability, aided by the minimal fuel cell run hours and hydrogen consumption. With solar providing most of the generation, emissions are extremely low at 3.47 kg/year of nitrogen oxides. The hydrogen tank storage profile demonstrates a steady accumulation from January through April, reaching full capacity (~80 kg) by early May (Fig. 12). From May onward, the storage level is maintained consistently at maximum, indicating that the system effectively meets its hydrogen production goals

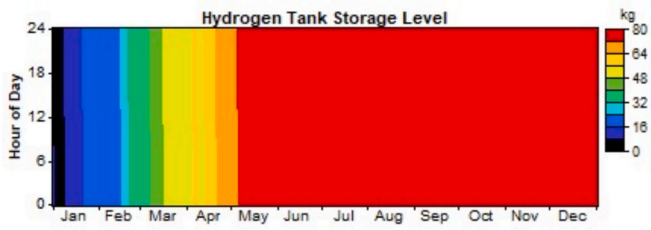


Fig. 12. Annual hourly hydrogen availability for the energy system for Paternoste.

and maintains storage availability for potential use across various seasonal conditions. This pattern reflects strong electrolyzer performance and consistent renewable energy supply during the operational year.

The power dynamics between the three principal components of the Ramsgate system during a typical June week are captured in Fig. 13. Ramsgate features the largest PV array (25 kW) among all sites, evident in the higher solar generation peaks reaching 20–21 kW during optimal conditions. Despite having more PV capacity, the system still requires moderate fuel cell support, particularly during evening hours and around hours 45–50 and 150–165, with fuel cell outputs of 8–10 kW. The AC Primary Load maintains a consistent pattern between 6–8 kW with occasional peaks approaching 10 kW. Ramsgate's operational pattern shows significant excess electricity generation (4,537 kWh/yr, the highest among all sites), which helps compensate for the location's lower annual solar radiation. The fuel cell operates at a capacity factor of 0.590 %, producing 1,554 kWh annually. This configuration results in the highest total net present cost (\$138,239) and a relatively high levelized cost of energy (\$0.272/kWh) among the four locations. The substantial excess generation, however, provides greater system reliability and operational flexibility, which may be valuable for areas with more variable renewable resources.

The total project cost of \$215,755.4 is primarily driven by capital expenditure (76.5 %), followed by replacement costs (21.5 %) and operating expenses (2.1 %) (Fig. S10). Fuel costs remain negligible, while the salvage cost is \$77,516. The hybrid system shows high initial investment, primarily in the fuel cell (\$90,000), hydrogen tank (\$31,000), and PV system (\$29,500) with scheduled replacements occurring for the battery, converter, and electrolyzer (Fig. S1).

Against diesel-only operation, Ramsgate shows the most modest economic gains with a present worth of \$75,338, annual worth of \$3,387/yr, return on investment of 6.77 %, internal rate of return of 3.55 %, and the longest simple payback period of 23.3 years, though still indicating measurable financial benefits from transitioning to hybrid renewable energy systems (Table 5).

The Ramsgate sensitivity analysis (Fig. S12) examines three competing system configurations across FC Capital Multiplier and FC Replacement Multiplier ranges from 0.0 to 1.0: PV/FC/Electrolyzer, PV/FC/Gen/Electrolyzer/Battery, and PV/Gen/Battery. The plot reveals a small triangular region in the lower-left corner where the PV/FC/Electrolyzer system becomes optimal when both cost multipliers remain

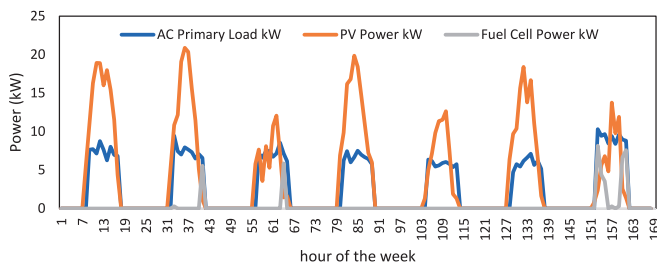


Fig. 13. Operational dynamics of the hybrid renewable energy system in Ramsgate.

below approximately 0.10 to 0.15, occupying less than 5 percent of the parameter space. Adjacent to this is a transitional zone where PV/FC/Gen/Electrolyzer/Battery proves economical when capital multipliers stay below 0.30 and replacement multipliers below 0.40, covering roughly 20 percent of scenarios. The PV/Gen/Battery configuration dominates approximately 75 percent of the parameter space, becoming optimal when either multiplier exceeds 0.30. The dual-boundary pattern indicates that hydrogen-only systems require exceptionally low costs (below 15 percent of baseline), while hydrogen-battery hybrids extend viability to moderate cost reductions. These findings suggest that Ramsgate's deployment strategy should prioritize conventional battery-diesel systems unless fuel cell costs achieve substantial reductions exceeding 60 to 70 percent below current values.

Reliability assessment of the configurations

The reliability performance of the four hybrid energy system configurations (Alexandra Bay, Jeffery Bay, Paternoster, and Ramsgate) was analyzed using calculated values for ENS and LPSP (Fig. 14). Results show that all configurations achieved exceptional essential load reliability, meeting desalination energy requirements with minimal interruptions. Alexandra Bay and Ramsgate demonstrated perfect essential load reliability at 100 %, with zero ENS and LPSP of 0.00, indicating uninterrupted desalination operations throughout the year. Jeffery Bay achieved near-perfect performance with 99.14 % reliability (ENS: 196 kWh/year, LPSP: 0.0086), closely followed by Paternoster at 99.18 % (ENS: 187 kWh/year, LPSP: 0.0082). The Primary Load Ratio (PLR), which measures energy distribution between desalination and hydrogen production, showed Alexandra Bay at 78.88 %, Jeffery Bay at 78.42 %, Ramsgate at 75.02 %, and Paternoster at 71.52 %. The results indicate that while all configurations maintain excellent desalination reliability (exceeding 99 %), PLR variations reflect differences in electrolyzer utilization patterns rather than service quality degradation. Configurations employing load-following dispatch strategy (Alexandra Bay, Jeffery Bay, Ramsgate) demonstrated both high essential load reliability and favorable PLR values, whereas Paternoster's cycle-charging strategy resulted in lower PLR due to increased electrolyzer operation (9,025 kWh/year), though maintaining comparable 99.18 % essential load reliability. This trend underscores the importance of distinguishing between energy allocation metrics (PLR) and actual reliability metrics (ENS, LPSP) in hybrid systems, particularly where hydrogen production operates as flexible, deferrable load. These findings demonstrate that all four coastal locations can support reliable renewable-powered desalination (reliability greater than 99 %), with performance differences primarily attributable to solar resource variations and operational strategies rather than fundamental technical limitations, providing a strong foundation for deployment planning across South Africa's coastal regions.

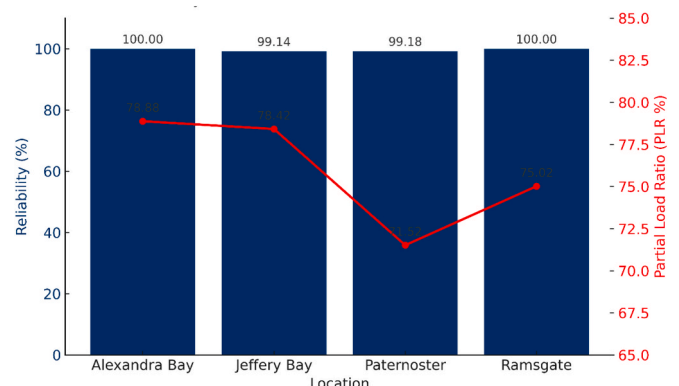


Fig. 14. Results of Energy not served and loss of power supply probability.

Water consumption analysis

Water consumption used in hydrogen production was evaluated to assess the tradeoff between environmental burden and hybrid renewable energy systems related to the addition of hydrogen storage system. Water use was calculated from annual energy input to the electrolyzer subsystem with a water use ratio of 0.9 L/kWh based on existing literature]. This rate includes both the electrochemical demands and other auxiliary consumption such as humidification and cooling. Fig. 15 shows the annual water demand at the four sites. The energy system configuration for Pat showed the highest water demand, around 8,122.5 L/year due to its increased electrolyzer load of 9,025 kWh/year. Ram is ranked second with a consumption of 6,846.3 L a year and Jeff and Alex consumed 5,608.8 and 5,504.4 L per year, respectively. These differences are a consequence of the electrolyzer load served, corresponding to the operational dynamics and dispatch strategies of each configuration. Specifically, the cycle-charging (CC) strategy implemented in the Pat system appears to have led to increased hydrogen production events and consequently higher water consumption. Conversely, load-following (LF) such as that carried implemented in Alex and Jeff resulted in lower electrolyzer utilization and more conservative water footprints.

From a sustainability standpoint, the results demonstrate a tradeoff of serious concern. Whilst hydrogen storage contributes to self-sufficiency and reliability of energy in renewable energy systems, it places a significant water burden, particularly in water-stressed or arid regions. This is especially important in off-grid or remote installations where water supply infrastructure may be limited or seasonal. Moreover, incorporating water consumption as an environmental criterion in the decision-making framework allows for a more holistic system evaluation. High-performing systems in terms of cost or reliability may underperform when assessed through a water sustainability lens. For example, although the Pat system delivered high hydrogen productivity, its elevated water demand may render it unsuitable for deployment in regions where water-energy nexus constraints are binding. Therefore, water consumption should be treated as a core environmental performance indicator alongside greenhouse gas emissions, NO_x output, and energy efficiency. In subsequent stages of the study, this metric was included as a cost-type criterion in the Multi-Criteria Decision-Making (MCDM) framework, ensuring that system configurations are ranked not only by economic and technical performance but also by their environmental footprint and resource sustainability.

MCDM

The analysis revealed significant variations in criterion importance, with environmental and operational factors dominating the decision framework. NO_x emissions emerged as the most influential factor,

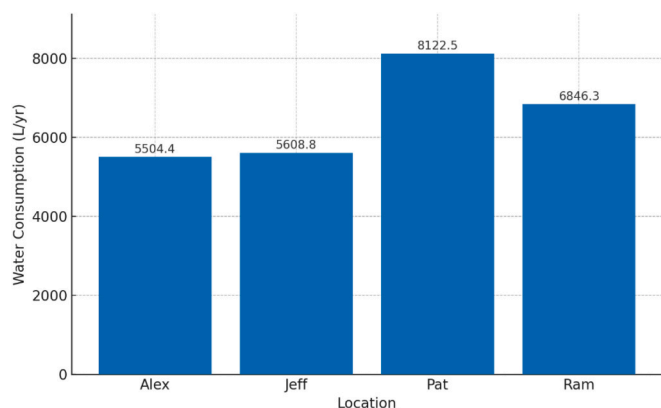


Fig. 15. Annual water demand for operating hydrogen energy storage at the four sites.

carrying a weight of 40.99 %, indicating that variations in emission performance across sites have the greatest impact on overall system evaluation (Fig. 16). Fuel cell production ranked second at 36.89 %, underscoring its central role in ensuring energy security, backup functionality, and system balancing within hybrid renewable configurations. Water consumption followed with 12.80 %, confirming the growing relevance of water-energy nexus considerations in sustainable system design. Together, these top three criteria account for 90.68 % of the total decision importance, demonstrating that environmental integrity and hydrogen utilization are the primary drivers of differentiation among the analyzed sites. Technological and economic indicators such as battery life (2.98 %), PV production (2.37 %), COE (1.08 %), total NPC (1.08 %), and water cost (1.07 %) exerted minimal influence, reflecting the relative maturity and convergence of hybrid renewable energy economics. Similarly, PLR (0.75 %) received the lowest weight, suggesting that reliability levels are uniformly high across all sites and therefore less discriminative. This weighting structure confirms that environmental performance and hydrogen-based operational resilience have overtaken cost metrics as the dominant determinants of system preference. The findings align with current sustainability paradigms emphasizing emission reduction and energy-water integration as strategic priorities. Consequently, these objective entropy-based weights validate that cleaner, low-emission configurations offer greater decision value and justify continued policy and investment emphasis on environmentally superior renewable-hydrogen hybrid systems.

To identify the most suitable location for system deployment, four Multi-Criteria Decision-Making (MCDM) methods (WSM, WASPAS, EDAS, and ARAS) were applied using entropy-derived weights. These methods incorporated both benefit and cost-type criteria, including environmental impact (NO_x emissions, water consumption), technical performance (fuel cell production, battery life, PV production), cost metrics (water cost, COE, NPC), and load distribution (PLR). Across all methods, Alexandra Bay consistently ranked first, demonstrating superior performance in terms of low NO_x emissions (3.47 kg/yr), minimal water consumption (5,504 L/yr), lowest fuel cell production (997 kWh/yr), and favorable economic metrics. Specifically, it achieved the top score under each model: WSM (0.9922), WASPAS (0.9914), EDAS (1.0000), and ARAS (0.9989) (Table 6). These results confirm that Alexandra Bay offers a balanced optimization of both economic and environmental criteria. Ramsgate followed as the second-best alternative, benefiting from moderate scores across most criteria, especially in NO_x emissions and fuel cell production metrics. Jeffery Bay consistently ranked third, while Paternoster ranked lowest across all four methods, largely due to its higher NO_x emissions (8.20 kg/yr), fuel cell production (2,210 kWh/yr), and water consumption (8,123 L/yr). The convergence of rankings across these distinct MCDM techniques reinforces the robustness of the decision framework. Notably, EDAS and ARAS, methods that explicitly incorporate positive and negative deviations, offered greater contrast among alternatives, which can be useful for sensitivity analyses.

To assess the consistency and reliability of the Multi-Criteria Decision-Making (MCDM) results, a Pearson correlation analysis was conducted on the final scores derived from WSM, WASPAS, EDAS, and ARAS methods. The results revealed a very strong and statistically significant correlation across all methods, with coefficients exceeding 0.94 in all pairwise comparisons. A heatmap visualization of the correlation matrix is provided in Fig. 17 to further illustrate the inter-method relationships. The highest correlation was observed between WSM and EDAS ($r = 0.999$), reflecting their shared emphasis on additive aggregation principles, with 99.75 % shared variance. Similarly, WASPAS and ARAS exhibited an extremely high correlation ($r = 0.996$), underscoring their complementary treatment of both additive and ratio-based performance evaluation. The correlation between methods from different MCDM methods (WASPAS, EDAS, WSM and ARAS) also remained very high ranging between 0.941 and 1.000. These values confirm the stability of the ranking outcomes across varying methodological

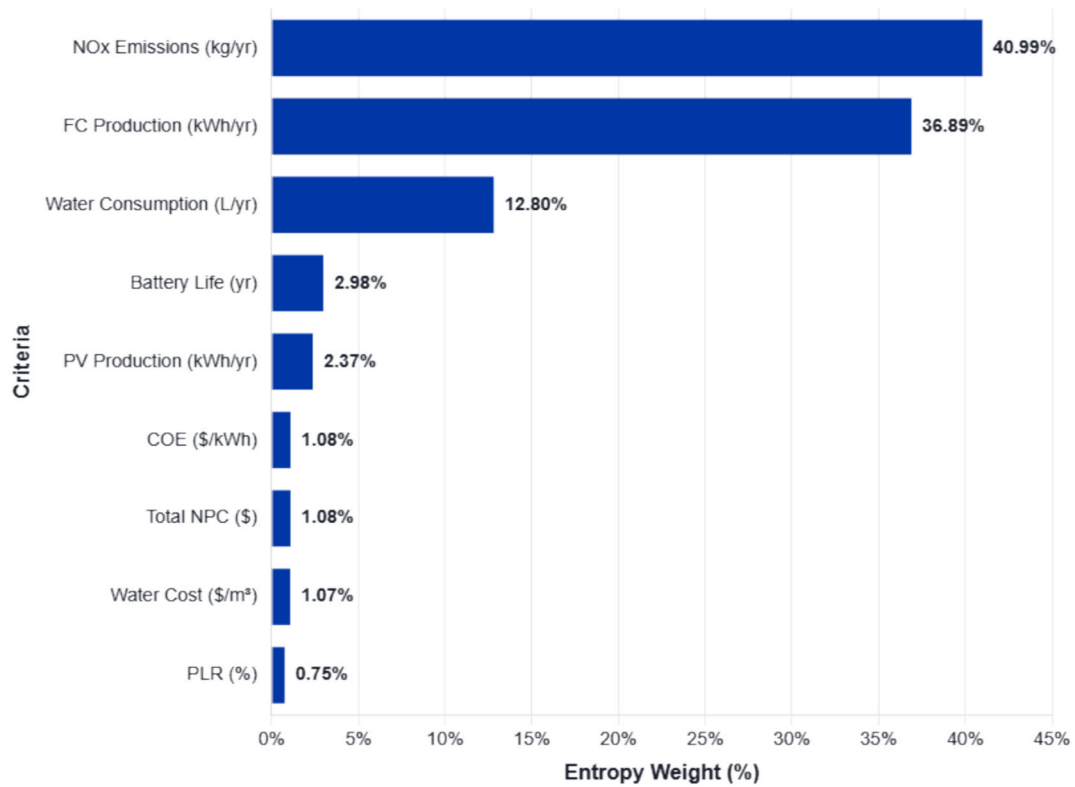


Fig. 16. Entropy weights for the selected criteria.

Table 6

MCDM scores and ranks for the 4 locations considered.

Location	WSM Score	WSM Rank	WASPAS Score	WASPAS Rank	EDAS Score	EDAS Rank	ARAS Score	ARAS Rank	Average Rank
Alexandra Bay	0.9922	1st	0.9914	1st	1	1st	0.9989	1st	1
Ramsgate	0.4994	2nd	0.2497	2nd	0.5277	2nd	0.6568	2nd	2
Jeffery Bay	0.4113	3rd	0.2057	3rd	0.412	3rd	0.6171	3rd	3
Paternoster	0.0371	4th	0.0185	4th	0.001	4th	0.4976	4th	4

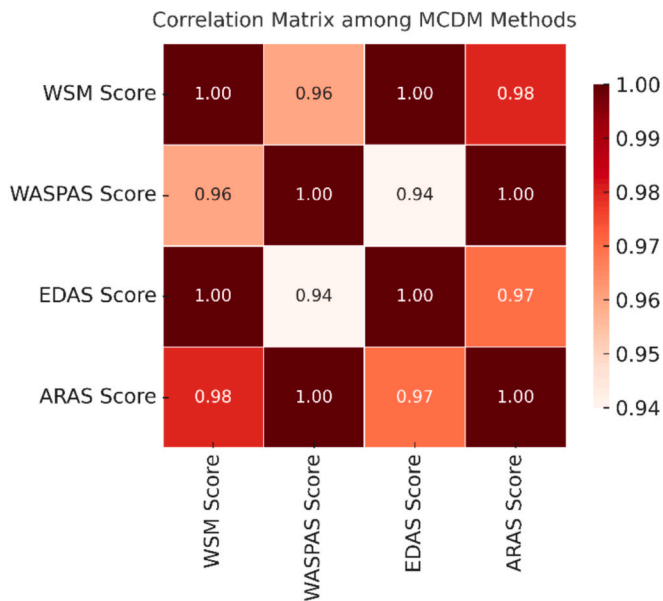


Fig. 17. Correlation matrix showing pairwise correlations between the various MCDM methods.

frameworks. The average pairwise correlation of 0.972 demonstrates exceptional consistency in alternative evaluation, with all methods converging on identical rankings (Alexandra Bay first, Ramsgate second, Jeffery Bay third, Paternoster fourth). This strong agreement among all four methods reinforces the robustness of the prioritization process and suggests that the top-ranked alternative (Alexandra Bay) holds a dominant position regardless of the decision model applied. These findings justify the use of multiple MCDM models in decision-making scenarios, as their convergence provides additional confidence in the final ranking outcomes.

Conclusions

This study evaluated the techno-economic, environmental, and reliability performance of hybrid renewable energy systems integrating solar PV, fuel cells, battery storage, electrolyzers, and hydrogen tanks for powering seawater desalination plants in four South African coastal locations: Alexandra Bay, Jeffery Bay, Paternoster, and Ramsgate. A combined simulation and multi-criteria assessment approach was employed, using HOMER Pro and entropy-weighted WSM, WASPAS, EDAS, and ARAS methods. The results show that Alexandra Bay provides the most favorable conditions for system deployment, achieving the lowest total net present cost of \$122,742, the lowest levelized cost of energy at \$0.242/kWh, and the lowest levelized cost of water at \$0.796/m³. Across all locations, the hybrid configurations demonstrated high reliability, with Alexandra Bay and Ramsgate achieving 100 % essential-

load supply and Jeffery Bay and Paternoster exceeding 99 %. Specific energy consumption remained consistent at 3.27–3.29 kWh/m³, aligning with values reported for reverse-osmosis desalination systems in similar studies. Solar PV contributed most of the total electricity generation, particularly in Alexandra Bay where PV provided 34,406 kWh/year (97 %), supported by hydrogen and battery storage during periods of low solar availability. Environmental results indicate that Alexandra Bay achieved the lowest NO_x emissions (3.47 kg/year) and lowest water consumption for hydrogen production (5,504 L/year), while Paternoster recorded higher values due to increased electrolyzer utilization. The load-following dispatch strategy used at Alexandra Bay, Jeffery Bay, and Ramsgate showed favorable performance in lowering emissions and reducing water consumption. Entropy-based weighting identified NO_x emissions, fuel cell production, and water consumption as the most influential criteria in ranking the system alternatives. Alexandra Bay ranked first across all MCDM methods, confirming its overall techno-economic and environmental superiority.

The findings demonstrate that renewable-hydrogen hybrid systems can provide reliable and sustainable power for desalination in South Africa's coastal regions, which continue to face significant water scarcity challenges. The study also offers a structured framework for evaluating future system designs and highlights the importance of integrating environmental metrics, cost parameters, and operational characteristics when selecting optimal configurations. Although the proposed hybrid systems showed high reliability across all locations, the consistently low fuel cell utilization and limited hydrogen use indicate that the installed hydrogen capacity may exceed the actual operational requirement. This pattern suggests that other system configurations, including PV battery options with higher solar capture or alternative storage sizing, may achieve similar reliability with lower overall cost. Future studies should examine a wider range of configurations under the same reliability conditions to clarify the circumstances in which hydrogen storage offers a measurable operational benefit. Further research may also extend this analysis by examining long-term variability in resource availability, exploring broader system configurations, and assessing large-scale deployment potential across additional coastal regions. Such efforts can support the development of sustainable water supply solutions and contribute to national objectives related to energy security, climate resilience, and Sustainable Development Goals.

Declaration of generative AI

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with the organization and language refinement of the paper. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the accuracy and integrity of the work.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecmx.2025.101498>

Data availability

Data will be made available on request.

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